EXPERIMENT 2: PROBLEM IDENTIFICATION & FEASIBILITY ANALYSIS

Course: Software Design Principles (SDP)
Class / Section: IT-B (2026-27)
Project Title: HOLD MONEY (Personal Finance System)
Repository: https://github.com/kenjaljedia/hold

TEAM COMPOSITION & MODULES

NAME	ROLL NUMBER	MODULE RESPONSIBILITY
Kenjal Jedia (Team Leader)	202401100600101	Application Core, Security Architecture & Integration
Ishan Gupta	202401100600092	Ledger Engine & Hold Allocation Logic
Garv Aditya Shandilya	202401100600077	Transaction Ingestion & Expense Categorization
Rudraksh Sharma	202401100600147	Analytics Dashboard, UI & CSV Data Export Module

AIM

To identify the problem addressed by the **HOLD MONEY** personal finance management application, formulate its formal problem statement, define clear objectives, establish project scope boundaries, and conduct technical and operational feasibility analyses.

OBJECTIVES

- Identify practical issues in existing financial applications regarding impulse spending.
- Draft a clear problem statement and objective baseline for reserve hold systems.
- Define project scope parameters including in-scope and out-of-scope functional features.
- Evaluate technical and operational feasibility for software deployment.

INTRODUCTION

Problem identification is the first step in software engineering. A well-defined problem statement and feasibility study ensure that the proposed software solution is practical, achievable, and aligned with user expectations before development starts. For financial management software, feasibility must consider double-entry ledger security, transactional consistency, client data privacy, and state persistence.

SELECTED PROJECT

Project Title: HOLD MONEY — Smart Expense Tracking & Reserve Management System
Domain: Fintech / Personal Finance Software Architecture

PROBLEM STATEMENT

Modern consumers struggle with effective budget discipline. Existing personal expense trackers are passive—they record spending after it occurs without preventing impulsive overdrafts or unbudgeted purchases. Manual envelope budgeting is tedious, while automated banking apps lack flexible user-defined temporary "holds" or virtual target reserves before actual settlement. Consequently, users frequently overspend, miscalculate disposable income, and fail to hit monthly savings goals.

The **HOLD MONEY** application resolves this by providing a smart financial control platform. It allows users to register accounts, track balance states, partition liquid funds into locked or temporary "Hold Balances" (earmarked for upcoming bills, savings goals, or recurring subscriptions), ingest transaction streams, and visualize real-time available liquidity versus total net worth.

PROJECT SCOPE

In-Scope: User account registration/login, liquid balance tracking, dynamic "Hold Vault" creation and release workflows, categorization of transaction debits/credits, analytics charts, encrypted data persistence, and CSV file data export.
Out-of-Scope (v1.0): Direct SWIFT/ACH wire settlements with commercial banks, credit score monitoring, stock market investment execution, and multi-signature corporate vault management.

TECHNICAL FEASIBILITY ANALYSIS

FACTOR	ASSESSMENT	TECHNICAL REMARKS
Technology Stack	Feasible	Python/JavaScript runtime, modern web UI, Git and GitHub for version control.
Ledger Engine	Feasible	Relational transactional data models enforce strict zero-sum double-entry rules.
Hold Allocation Logic	Feasible	Deterministic rules compute real-time available liquidity: *(Available = Total − Holds)*.
Security & Auth	Feasible	Hashed passwords (bcrypt) and session bearer tokens protect user data integrity.
Data Export	Feasible	Client-side CSV generator formats ledger entries without external library overhead.

OPERATIONAL FEASIBILITY ANALYSIS

FACTOR	ASSESSMENT	OPERATIONAL REMARKS
User Acceptance	High	Directly addresses user impulse spending via visual and logical hold allocations.
Ease of Use	High	Clean, intuitive dashboard requires zero accounting training to navigate.
Behavioral Impact	High	Visualizing earmarked reserve holds deters users from spending vital bill money.
Maintainability	High	Modular code layout cleanly separates UI, business logic, and ledger storage.

OBSERVATION & ANALYSIS

The feasibility analysis confirms that **HOLD MONEY** is technically viable and operationally feasible. Technical dependencies utilize open-source frameworks, and risk mitigations (such as transaction concurrency safeguards) ensure system stability during operational execution.

RESULT

The problem statement, project scope, detailed objectives, and technical and operational feasibility analysis for the **HOLD MONEY** system were successfully prepared.